big civilian aircraft that could break the sound barrier remained a pipedream for the US. Aerospatiale-BAC flew the first Concorde as early as 1969, and entered service seven years later. The US tried hard to develop and fly its equivalent, but in vain. The French success with Concorde was so difficult for the US to digest that it banned Concorde from its airspace after many years of trans-Atlantic flights by Air France and British Airways. The US gave some safety concerns as the reason, but within, its spin doctors knew what the consequence would be. Without flying across the Atlantic, a supersonic civilian plane cannot be viable. Concorde, which had twenty planes and one crash (in 2000) in its twenty-seven years of flying, had its last ceremonial flight in 2003.

It was this confidence that drove France to plan a geosynchronous satellite launch vehicle in the early 1970s. And it presented us with an opportunity to further our agenda. It was an ambitious project at that time. The three-stage vehicle called LIIIS would be an all-liquid fuel vehicle, including a cryogenic upper stage.

Configuring a satellite-launching rocket is a top-down process, literally, starting from the payload under the nose cone and planning down to the first stage engine and boosters. The first thing to decide is the orbit—you have to be clear on where in space you want to place the satellite. That depends on what you want the satellite to do, which again decides what kind of a payload you need. If it is for communication, you need transponders; earth-watching satellites need cameras.

The weight of the payload decides the configuration of the rocket as there should be optimal balance of the payload and propellants. The stages of the rocket—solid, liquid or cryogenic—are dictated by this balance.

While India continued its romance with solid propellants ignoring the merits of liquid fuels, the French were among the earliest to plan a geosynchronous satellite launch vehicle powered entirely by liquid fuel, given that cryogenic engine, too, deals with liquefied gases. For a primer, here is a quick list of advantages of liquid over solid: a liquid engine can be tested repeatedly before being flown using the same hardware after cleaning and reassembly, enhancing reliability; a solid engine burns itself out, which